Experiment 13: Hill Cipher Known-Plaintext Attack:

```
import numpy as np

P = np.array([[7, 4],
              [8, 11]])     # Plaintext matrix

C = np.array([[19, 2],
              [2, 13]])     # Ciphertext matrix

P_inv = np.array([[25, 22],
                  [22, 23]]) # Inverse of P mod 26

K = np.dot(C, P_inv) % 26

print("Plaintext Matrix:")
print(P)

print("\nCiphertext Matrix:")
print(C)

print("\nRecovered Key Matrix:")
print(K)
```

Output:

```
Plaintext Matrix:
[[ 7  4]
 [ 8 11]]

Ciphertext Matrix:
[[19  2]
 [ 2 13]]

Recovered Key Matrix:
[[25 22]
 [24  5]]

=== Code Execution Successful ===
```